

Lecture 24

⇒ Applications of eigenvalues

$$A = \begin{bmatrix} 0.1 & 0.01 & 0.3 \\ 0.2 & 0.99 & 0.3 \\ 0.7 & 0 & 0.4 \end{bmatrix} \Rightarrow \text{markov matrix.}$$

① $a_{ij} \geq 0$

② $\sum_{i=0}^{\infty} a_{ik} = 1$

In case of difference equations, the steady state is defined when $\lambda = 1$

⇒ Somehow the second property guarantees that the markov matrices will have an eigenvalue of 1.

- ① $\lambda = 1$ is an eigenvalue
- ② All other $|\lambda_i| < 1$
- ③ $x_i \geq 0$

$$\Rightarrow u_k = A^k u_0 = c_1 \lambda_1^k x_1 + c_2 \lambda_2^k x_2 + \dots$$

$$\# (A - 1I) = \begin{bmatrix} -0.9 & 0.1 & 0.3 \\ 0.2 & -0.1 & 0.3 \\ 0.7 & 0 & -0.6 \end{bmatrix}$$

Now for this matrix, we can see that all columns add to zero.

$\Rightarrow$ We can somehow show that the columns are dependent.

$$\Rightarrow \boxed{A_1 + A_2 + A_3 = 0}$$

↓
(rows)

$\Rightarrow$ So they have dependent rows
hence $\boxed{\text{rank}(A) < n}$

$\Rightarrow$ This also tells us that
for square matrix, it would
be singular.

$\Rightarrow$ Hence $\det(A - I) = 0$

So, 1 is always an eigenvalue

$\Rightarrow x_1$ is in the nullspace of
 $(A - I)$

Fact

$\boxed{\text{eigenvalues of } A = A^T}$

$$\begin{aligned} Ax_0 &= \lambda x_0 \\ x_0^T A^T x_0 &= \lambda x_0^T (x_0) \\ x_0^T (\underbrace{A^T x_0}) &= x_0^T (\underbrace{\lambda x_0}) \end{aligned}$$

$$\begin{aligned} \det(A - \lambda I) &= 0 \\ \det(A^T - \lambda I^T) &= 0 \\ \det(A^T - \lambda I) &= 0 \end{aligned}$$

Markov matrices

⇒ Suppose we are looking at the system involving two cities and movement of people between them and we want to see how to model it.

→ It can be modeled as a Markov matrix as the total population remains the same and a fraction stays and a fraction goes.

$$\begin{array}{c} \text{population in city 1} \\ \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}_{t=k+1} = \begin{bmatrix} .9 & .8 \\ .1 & .2 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}_{t=k} \\ \downarrow \quad \downarrow \\ \text{population in city 2} \quad \boxed{\text{city 1}} \end{array}$$

Let's say for this particular example, everyone started in city 2.

$$u_0 = \begin{bmatrix} 0 \\ 100 \end{bmatrix}$$

$$u_{k+1} = \begin{bmatrix} .9 & .8 \\ .1 & .2 \end{bmatrix} u_k$$

$$\begin{bmatrix} -0.1 & 0.8 \\ 0.1 & -0.2 \end{bmatrix} \quad \underline{\lambda = 1}, \quad \underline{0.1}$$

$$x_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$x_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$u_k = c_1 (1)^k \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 (0.1)^k \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$u_k = c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 (0.1)^k \begin{bmatrix} -1 \\ 1 \end{bmatrix} \quad \boxed{k=0}$$

$$c_1 - c_2 = 0$$

$$c_1 + c_2 = 100$$

$$u_k = 50 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \frac{50}{(10)^k} \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

Projections with orthonormal basis

$$v = x_1 q_1 + x_2 q_2 \dots x_n q_n$$

→ Nice formula to find a particular q_k

$$v \cdot q_k = (x_1 q_1 \dots x_n q_n) \cdot q_k$$

$$v \cdot q_k = x_1 (q_k \cdot q_k)$$

$$v \cdot q_k = x_1 \|q_k\|^2 = 1$$

$$v \cdot q_k = x_1$$

$$x_k = q_k^T \cdot v$$

$$Qx = v$$

$$x = Q^{-1} v$$

$$x = Q^T v !$$

because

$$Q^{-1} = Q^T$$

for orthonormal

Fourier Series

$$f(x) = a_0 + a_1 \cos x + b_1 \sin x + a_2 \cos 2x + b_2 \sin 2x \dots$$

$$\Rightarrow v^T w = v_1 w_1 + v_2 w_2 \dots + v_n w_n$$

$\Rightarrow$ for functions

$$f^T g = \int f(x) g(x) dx$$

$\Rightarrow$ For Fourier series, the limits would be $0 \rightarrow 2\pi$

- Let's check this integral for all the terms in Fourier series

$$\Rightarrow \int_0^{2\pi} \sin x \cos x \, dx$$

$$= \frac{1}{2} (\sin x)^2 \Big|_0^{2\pi}$$

And we can prove the same for all $\sin(nx) \cos(mx)$, $\sin(nx) \sin(mx)$, $\cos(nx) \cos(mx)$ pairs.

$\Rightarrow$ Now all the terms in the Fourier series are sort of orthonormal vectors in a continuous space.

Now using the projection thing we found out,

$$V = x_1 q_1 + x_2 q_2 \quad \dots \quad x_n q_n$$

$$f(x) = a_0 + a_1 \sin x + a_2 \cos x \dots$$

⇒ Basically this vector

$$\begin{bmatrix} 1 \\ \sin x \\ \cos x \\ \sin 2x \\ \cos 2x \\ \vdots \end{bmatrix}$$

spans the whole
function space!!!

⇒ To find a particular constant.

$$\Rightarrow \int_0^{2\pi} f(x) \cos x \, dx = a, \quad \int_0^{2\pi} (\cos x)^2 \, dx$$

$$a_1 = \frac{\int_0^{2\pi} f(x) \cos x \, dx}{\pi}$$